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Navigating career stages in the age of artificial intelligence: A systematic interdisciplinary review and agenda for future research

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ABSTRACT

As artificial intelligence (AI) use expands within organizations, its influence is increasingly permeating careers and vocational domains. However, there is a notable lack of structured insights regarding AI's role in shaping individual career paths across career stages. To address this gap, we undertook a systematic literature review of 104 empirical articles, aiming to synthesize the scholarship on AI in the context of careers. Drawing upon career stage theory, we examine the implications of AI on careers, identify key barriers and enablers of AI use in this area, and reveal how the utilization of AI impacts individuals' career competencies. In doing so, we illustrate how AI actively shapes individuals' career trajectories and we dissect these effects both within and across various career stages to situate AI within the broader context of careers research. Adopting a sustainable career lens, we conclude by outlining a future research agenda that advocates for the design and adoption of AI systems that promote sustainable and equitable careers.

1. Introduction

Artificial intelligence (AI) is revolutionizing many aspects of life and is an increasingly important feature of organizations. AI refers to "a collection of interrelated technologies used to solve problems and perform tasks that, when humans do them, requires thinking" (Walsh et al., 2019, p. 14).[1] The most recent PwC Global CEO survey (2024) highlights the potential competitive advantage that AI offers to organizations, with 70 % of global CEOs believing that AI will significantly re-shape how their organizations create, deliver, and capture value in the next three years (PwC, 2024). However, it is critical to recognize the significant investments and potential risks associated with AI use, underscoring the need for careful analysis of its impacts and outcomes (Lee et al., 2023), including how AI integration influences individuals' working lives (Bankins et al., 2024).

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[1] Throughout the paper, we use the terms 'AI' and 'AI systems' interchangeably, recognizing that AI refers to a broad class of technologies and that there are many different types of AI.

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In recent years, the discourse surrounding AI has extended to careers and vocational domains (Baruch & Sullivan, 2022). Forging this connection is important, as the wide-ranging applications of AI manifest in multiple ways that can influence careers. For example, AI is used as a tool for on-demand and personalized career guidance via AI-powered career advisors like Jobiri (European Innovation Council, 2023) and recommender systems that suggest optimal career changes (Dawson et al., 2021). The use of such systems has downstream implications for individuals' educational and career choices. Furthermore, the increasing integration of AI into organizations is re-shaping job designs and skill demands and requirements (Bankins et al., 2024), potentially transforming the nature and advancement pathways of various careers.

Yet amidst these organizational transformations, a critical oversight persists in the literature: the lack of a systematic exploration of the broader implications of AI's disruption, transformation, and amplification of individuals' careers. Although current discussions, both in popular and scholarly circles, examine how AI influences organizational contexts, existing investigations do not fully capture the expanding role of AI in career management (for a review, see Bagdadli & Gianecchini, 2019). While research has explored the consequences of AI on broader areas like human resource management (HRM) (see Table S1, supplementary material), there is a significant gap in understanding how AI shapes individuals' progression through various career stages. Given the growing concerns about the future of work and the potential career ramifications of technological advancements, it is imperative to investigate the role of AI in shaping career trajectories more thoroughly. Moreover, a systematic understanding of how AI influences careers affords evidence-based guidance for implementing AI technologies to support an equitable and inclusive job market. Doing so holds promise for fostering sustainable careers, which is particularly pertinent given the emerging ethical issues associated with AI and careers (Westman et al., 2021).

In our systematic literature review, we draw upon career stage theory (Super, 1957) to generate insights into these critical issues. A career is more than a chronological sequence of events; instead, a career is an amalgamation of experiential threads woven across time (Louis, 1980). A career stage perspective (Sullivan & Al Ariss, 2021) considers careers as a series of transitions, denoting both an objective role change, and a subjective change in perceptions of a role (Sullivan & Crocitto, 2007). We adopt a stage-based perspective centred on recognized transitional career stages, including *school* (i.e., primary and high school students), *university-to-work* (i.e., university students); and *established workers* (i.e., those currently in employment), which may encompass various inter- and intra-role transitions (see Louis, 1980; Krau, 1983; Nicholson & Arnold, 1989; Super, 1957). In addition, we explore the career competencies developed by individuals through their utilization of, or interaction with, AI. Career competencies consist of "knowledge, skills, and abilities central to career development, which can be influenced and developed by the individual" (Akkermans et al., 2013, p. 249). These competencies differ from specific job-related skills as career competencies can be honed over time (Akkermans et al., 2013) and are essential for effective career management across the lifespan (Barnes et al., 2022).

In sum, we address the following research questions: RQ1: What are the implications of AI within and across career stages?; RQ2: What are the key barriers and enablers of utilizing AI within and across career stages?; RQ3: How does the use of AI impact individuals' career competencies within and across career stages?

By examining the link between AI and careers, we make several contributions to the literature. While we contribute to systematic investigations of the role of AI in the work context (e.g., Bankins et al., 2024; Lee et al., 2023; Pereira et al., 2023), our research particularly advances the growing discourse on how AI affects career outcomes specifically (see Muhammad, 2023; Pandya & Wang, 2024). We do so by providing a systematic and interdisciplinary literature review that synthesizes the rapidly expanding body of empirical knowledge concerning AI's influence across diverse career stages. Employing bibliometric and thematic analyses, we offer a comprehensive overview of the existing literature. This approach affords us a broad view of the field before delving into a more detailed assessment of how AI influences various career stages. Our primary contribution is the delineation of AI's active influence on individuals' career trajectories, dissecting these impacts both within and across career stages, thereby integrating AI into the broader context of careers research. Consistent with this focus, we also identify the barriers and enablers of AI use in careers, as well as its effects on career competencies. Finally, building on our review findings, we propose a sustainability-focused future research agenda (De Vos et al., 2020). This agenda centers on how AI can be leveraged to develop sustainable and equitable careers, thereby helping to promote healthy, fulfilling, and productive vocational experiences.

2. Literature search method and analytical approach

2.1. Search strategy

Fig. 1 illustrates our article search process, which was designed to ensure both precision and inclusivity in generating our sample of literature. Using the Scopus database, we adopted a two-pronged approach to create our search string. First, our search string incorporated terms related to technology, and those associated with careers and vocational services such as 'career development' and 'vocational guidance'. This initial search yielded a relatively modest compilation of fewer than 70 sources. To safeguard against sacrificing search breadth, we conducted an additional search using the term 'career' (rather than the more specific career-related terms). Second, to capture work from various disciplines, we included subject areas such as business, psychology, decision sciences, sociology, and economics, with a selective focus on journal articles. The second search string yielded a broader collection of papers ($n = 349$), which we then used as our primary search string. Our filtering process, in conjunction with predetermined inclusion and exclusion criteria (see Fig. 1), culminated in a final sample of $n = 104$ papers.

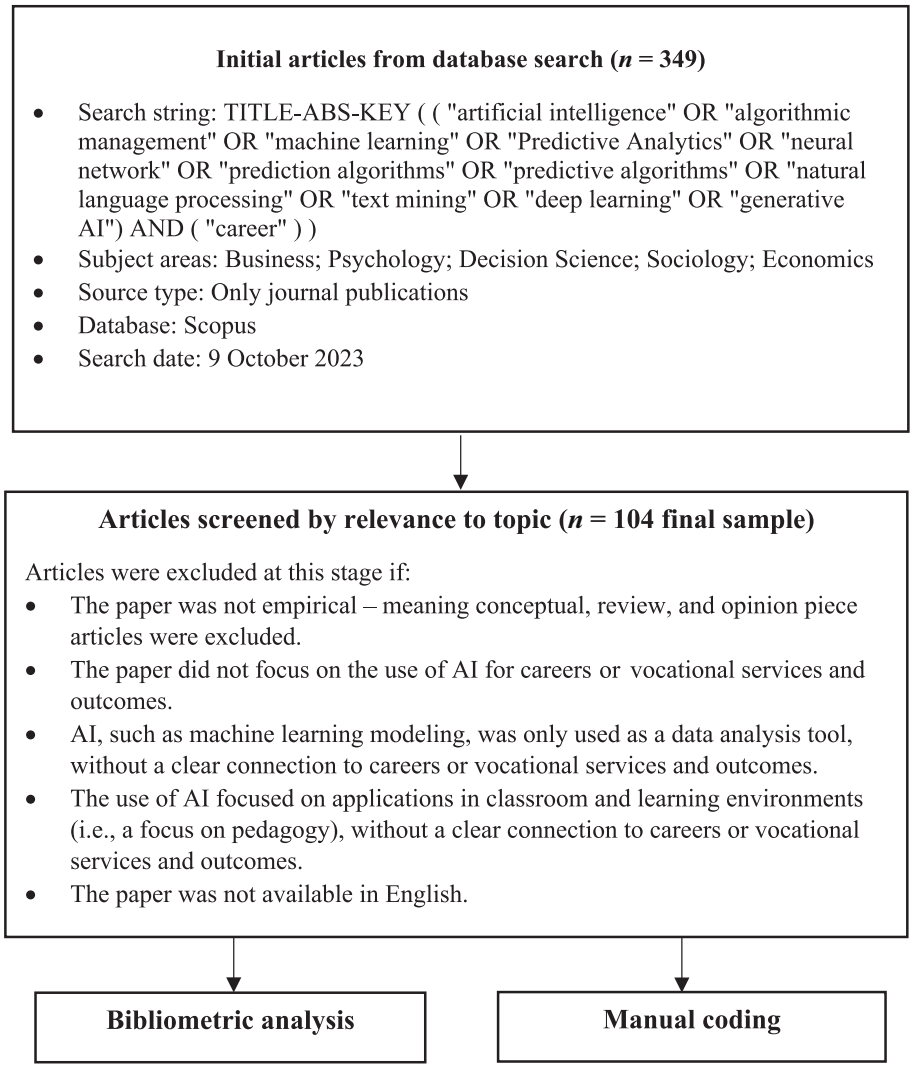

Fig. 1. Flow diagram of search process and inclusion criteria.

2.2. Analytical approach

Our analysis comprised two phases employing complementary methods. In the first phase, we employed bibliometric analysis, through bibliographic coupling, to identify and analyze clusters of research. This technique uses citation analysis to cluster research based on the extent of shared references within the reference lists of each source (Jarneving, 2007). The degree of reference overlap signifies thematic similarity, which generates clusters of research examining similar topics (Wu et al., 2021). For this phase, we used VosViewer, which removed isolated papers to generate meaningful and accurate clusters (Traag et al., 2019).

In the second phase, we utilized manual coding via immersion in the sample of literature (for a similar approach, see Bankins et al., 2024; Collins et al., 2023). This analysis allowed us to apply our career stages framework to organize the existing literature and to identify the implications of AI use within and across these stages. The Phase 2 analysis adopted a three-step process. First, we grouped our sample of papers into categories that reflected the three primary career stages we previously identified: school (i.e., primary and high school students); university-to-work (i.e., university students); and established workers (i.e., those in employment). Second, we developed a data extraction form to capture important descriptive details for each article, such as methods used, sample characteristics, and theories applied. To code these aspects, we trained six research assistants who were at advanced stages in their management degrees. Third, the data extraction form also captured more substantive content of the articles, aligned with our research questions, such as how AI is being used within each career stage, its impact on career and vocational services, barriers and enablers to AI use in each stage, and overall career outcomes. As this coding process was more detailed and required reading each article, it was conducted by the first author. This process was flexible to capture emergent themes. In the next sections, we present the results of the Phase 1 and Phase 2 analyses.

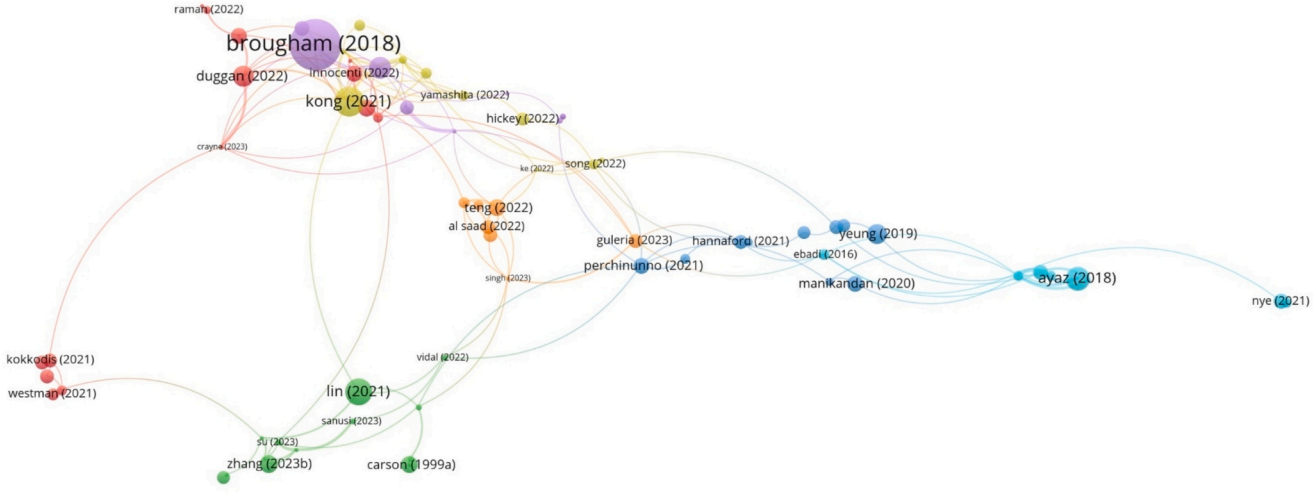

Fig. 2. Bibliographic coupling: visualization of the seven clusters of research.

3. Phase 1 findings: bibliometric analysis

The first phase of our analysis affords insights into the broad landscape of the AI and careers literature. Fig. 2 visually represents the seven clusters of research identified through bibliographic coupling. This clustering reveals that the current body of research on the utilization of AI in the careers domain is fragmented, characterized by small and disconnected clusters with limited cross-referencing between them. The absence of cross-referencing indicates a lack of integration among these research areas. Additionally, the analysis suggests that this body of work is not yet highly cited, as indicated by the relatively small size of the cluster circles. The most frequently cited paper in our sample (i.e., Brougham & Haar, 2018) focuses on established workers' perceptions of smart technology, artificial intelligence, robotics, and algorithms (STARA) in their workplaces. It identifies several negative outcomes associated with employees' STARA awareness.

Table 1 provides a descriptive overview of the seven clusters of research, which we briefly outline here. *Cluster 1* (red color circles) focuses on utilizing AI for career guidance, particularly for university students, by leveraging AI's predictive capabilities for activities such as recommending career paths (Kokkodis & Ipeirotis, 2021), matching students to courses (Alshaikh et al., 2021), and assessing employability (Raman & Pramod, 2021). *Cluster 2* (green color circles) centers on students and focuses on exploring ways to develop AI literacy (e.g., Sanusi et al., 2023; Zhang et al., 2023), with a smaller sub-set of work examining predictors of career choice. *Cluster 3* (dark blue color circles) revolves around university students and predicting their academic performance (Manikandan & Chinnadurai, 2020), alongside predictors of their entry into STEM (Science, Technology, Engineering, Mathematics) careers (Liu & Tan, 2020). *Cluster 4* (yellow color circles) investigates how AI is impacting established workers' careers, with a focus on career sustainability (Hussain et al., 2023), career self-management (Presbitero & Teng-Calleja, 2022), career competencies (Kong et al., 2021), and career perceptions (Zhang & Jin, 2023), often within the context of hospitality workers. *Cluster 5* (purple color circles) examines the use of AI for predicting performance indicators, such as academic achievement for university students (So et al., 2023) and the professional performance of established workers, such as lawyers (Lopes et al., 2018). Relatedly, *Cluster 6* (light blue color circles) investigates how AI can predict aspects of academic researchers' careers, including their impact, citation trends, and collaborative networks (Ayaz et al.,

Table 1
Overview of Phase 1 findings and analysis: bibliographic coupling.

Cluster number, label, color, and number of papers[a]	Cluster description	Example journals
1 – AI for career guidance (red; $n = 15$)	Focuses on using AI to provide career guidance through actions like: matching students to courses; assessing employability; predicting student performance; and recommender systems for career paths. There is a focus on university students, with some research on how AI is disrupting established workers in specific sectors, like audio mastering and law.	International Journal of Emerging Technologies in Learning; Journal of Learning Analytics; Sustainable Operations and Computers; International Journal of Human Resource Management
2 – Developing AI literacy (green; $n = 14$)	Focuses on developing students' AI literacy and how to best achieve this outcome across age ranges. A sub-section of work examines how AI's predictive capabilities can be applied to predicting students' career choices. There is a focus on primary and secondary school students, as well as university students.	Computers and Education: Artificial Intelligence; Education and Information Technologies; International Journal of Artificial Intelligence in Education; Journal of Computer Assisted Learning
3 – Predicting student performance and STEM career choice (dark blue; $n = 10$)	Focuses on predicting student performance and dropout and predicting the likelihood of students entering STEM careers. There is a focus on university students.	International Journal of Artificial Intelligence in Education; International Journal of Engineering Education; Journal of College Student Retention: Research, Theory and Practice; Journal of Educational Data Mining
4 – AI impacts on established workers' careers (yellow; $n = 10$)	Focuses on the impact of AI on careers generally, for example, in terms of career sustainability, career self-management, career competencies, and career perceptions. Several studies examine these aspects in the hospitality sector. Established workers are the focal samples.	International Journal of Contemporary Hospitality Management; Journal of Hospitality Marketing and Management; Journal of Vocational Behavior; Personnel Review
5 – AI and prediction of performance (purple; $n = 9$)	Research in this cluster ranges across utilizing the predictive capabilities of AI for predicting academic achievement (university students) and workers' performance (in sectors such as law and academia). Both university students and established workers are focused upon.	IEEE Transactions on Learning Technologies; International Journal of the Legal Profession; International Journal of Productivity and Performance Management; Journal of College Student Retention: Research, Theory and Practice
6 – AI and academic careers (light blue; $n = 9$)	Focuses on academia and research careers and how AI can be used to predict career-related factors, such as: the impact and publication of future research papers; citation trends; and the underpinnings of scientific networks and collaborations. Established workers are the focus of this cluster.	International Journal of Artificial Intelligence in Education; Journal of Research on Technology in Education; Scientometrics; Journal of the Association for Information Science and Technology
7 – AI in healthcare education and careers (orange; $n = 7$)	Focuses on university students' perceptions of AI in higher education broadly, but also more specifically in the medical and healthcare sectors and their associated programs of study. The focus is on university students.	BMC Medical Education; Education and Information Technologies; JMIR Medical Education; Open Public Health Journal

[a] The total number of papers in the bibliographic coupling (Phase 1 analysis) is less than the $n = 104$ final sample. This is because the process of entity linking determines whether papers are connected via their reference lists. Thirty papers in our final sample were determined not to be connected in this way and so were excluded from the bibliometric (clustering) analysis. However, the full $n = 104$ papers were manually coded (Phase 2 analysis).

2018; Zhou et al., 2022). *Cluster 7* (orange color circles) centers on AI in the healthcare and medical profession, along with university students' perceptions of AI's impact on their learning in higher education and subsequent careers in this profession (Pucchio et al., 2022; Teng et al., 2022).

Overall, these findings indicate that the extant literature has yet to form a cohesive and integrated foundation for advancing knowledge regarding the impact of AI within vocational and careers domains. From a career stage perspective, several clusters span multiple career stages, such as Cluster 1 (university students and established workers), Cluster 2 (school and university students), and Cluster 5 (university students and established workers). This is noteworthy since careers theory suggests that individuals' needs differ across these stages (e.g., Super & Hall, 1978). From an occupational perspective, certain occupational groups, such as those in the STEM fields of medicine and healthcare, and the hospitality sector, appear to be more actively engaged with AI's career implications for their workforces compared to other fields. This suggests that sharing insights generated within these occupational groups could prove valuable for others, fostering cross-pollination of knowledge. These Phase 1 findings afford an overarching picture of our focal literature.

4. Phase 2 findings: immersive, thematic coding by career stage

In Phase 2 of our analysis, we apply a theory-informed perspective based on career stages to organize and evaluate our literature sample. Careers theory recognizes that people enter different stages of their career and may cycle back through certain stages when they encounter disruptions or changes (Lent & Brown, 2020). For example, a recent graduate entering the workforce may choose to pursue further education to meet evolving job market demands. Each career stage signifies distinct individual needs at a given point in time, necessitating tailored vocational services and advice (Akkermans et al., 2024). Our Phase 1 analysis indicates that while the evaluation of AI's effects across different career stages is ongoing, it is unevenly distributed. By categorizing the literature into three transitional career stages (i.e., school; university-to-work; and established workers), we aim to anchor our review within an established framework for dissecting 'careers' across their primary stages (Louis, 1980; Krau, 1983; Nicholson & Arnold, 1989; Super, 1957). By focusing on multiple stages, this approach also aligns with contemporary calls to examine careers across the lifespan (e.g., Akkermans et al., 2024; Jiang et al., 2019). Consequently, our approach can generate deep insights into the current applications and outcomes of AI *within* each stage and make meaningful comparisons *across* these stages. Table 2 overviews the descriptive details of our sample of literature and outlines the key themes within each career stage. In the subsequent sections, we present our findings, structured around three core aspects that align with our overarching research questions.

4.1. The implications of AI at different career stages

4.1.1. School (i.e., primary and high school students)

This career stage is characterized by two main themes.

Using AI for career exploration and career guidance ($n = 9$). In this theme, several studies focus on leveraging AI to enable students to explore different careers and career paths and provide career-related guidance. These investigations show how chatbots and virtual mentors can fulfill various career-related purposes, including offering specific advice on university requirements and potential majors through virtual academic advisors (Assayed et al., 2023). AI was also employed to assist students in selecting courses, specializations, and career tracks based on their aptitudes and interests (Alshaikh et al., 2021; Guleria & Sood, 2023; Hernandez & Atienza, 2021). Machine learning techniques were also utilized to enable, scale, and diversify development opportunities through virtual mentors (Nye et al., 2021). Furthermore, AI's predictive capabilities were harnessed to forecast students' performance across both academic and non-academic factors, providing them with insights into how their performance impacts educational choices and future careers (Manikandan & Chinnadurai, 2020). These examples illustrate both distal career guidance (e.g., aiding students in making informed decisions about their educational paths leading to specific career trajectories) and proximal career guidance (e.g., providing access to mentors). While many AI models combined self-aspects of career exploration (related to personal interests and test scores) with external environmental aspects (related to aligning interests with specific occupational groups) (Xu et al., 2014), one study did examine how machine learning assessments of psychometric evaluations could enhance students' self-understanding to support career decision making (Singla et al., 2022).

Building students' AI literacy for their future career skill set ($n = 7$). This theme revolves around the development and evaluation of programs aimed at developing AI literacy, starting from preschool-aged children through to high school-aged young adults. AI literacy reflects a "set of competencies that enable individuals to critically evaluate AI technologies, communicate and collaborate effectively with AI, and use AI as a tool in various settings, including online, at home, and in the workplace" (Ng et al., 2023, p. 4). This research often positions the cultivation of AI literacy as an important component of students' future work skill set, equipping them to engage in fields and industries that rely on AI (Zhang et al., 2023) and to employ it in real-world contexts (Ng et al., 2023; Su & Yang, 2023). AI literacy programs encompass various aspects of the technology, generally combining technical components (such as developing a basic understanding of AI and its modeling) with social components (particularly concerning AI ethics and its societal impacts). For example, Zhang et al. (2023) outline three key domains underpinning the development of an "AI literate citizen": "age-appropriate" comprehension of technical concepts; an appreciation of the ethical and societal implications of AI; and an understanding of how AI impacts jobs and how individuals can enhance their career adaptability to manage such transitions. Studies in this theme also identify and assess different pedagogical approaches for developing AI literacy, including maker pedagogies (Ng et al., 2023) and ways to enhance students' motivation to learn about AI (Lin et al., 2021). Some articles suggest that AI literacy can also enhance other work-related skills, such as creativity (Hwang et al., 2022) and higher-order critical thinking abilities (Ng et al., 2023), and help students

Table 2
Overview of Phase 2 findings and analysis: descriptive details and overview of themes.

Career stage	Example journals	Themes in this career stage	Empirical methods used (% of papers in the theme)	Example theories/frameworks used
School, i.e., primary and high school ($n = 16$)	Computers and Education: Artificial Intelligence; Education and Information Technologies; Education Sciences; International Journal of Engineering Education; Journal of Vocational Behavior	(1) Using AI for *career exploration and career guidance* (2) Building students' *AI literacy* for their future career skill set	• Quantitative - 31 % • System/technology development - 38 % • Mixed/multiple methods - 31 %	AI/machine learning literacy; career assessment/guidance; career mentors; data mining and predictive analytics; explainable AI; sociotechnical systems
University-to-work, i.e., university students ($n = 49$)	Education and Information Technologies; Computers and Education; International Journal of Artificial Intelligence in Education; International Journal of Information and Communication Technology Education; JMIR Medical Education	(1) Using AI for *career guidance and academic advising* (2) Building students' *AI literacy* and *views on AI's impact on future careers* (3) Examining *job- or occupation-specific skills in AI-enabled workplaces* (4) Predicting *at-risk students and student attrition* using AI	• Quantitative - 37 % • System/technology development - 27 % • Qualitative - 12 % • Mixed/multiple methods - 18 % • Other - 6 %	Social cognitive career theory; precision education; signaling theory; skills taxonomies; teamwork competencies
Established workers, i.e., those in employment ($n = 39$)	Journal of Vocational Behavior; Journal of Hospitality Marketing and Management; Personnel Psychology; Scientometrics; International Journal of Human Resource Management	(1) Understanding the *career impact of AI* (2) AI *predictions of workers' performance and career trajectories* (3) AI-enhanced *career guidance* (4) *Skill needs* due to AI use	• Quantitative - 41 % • System/technology development - 26 % • Qualitative - 18 % • Mixed/multiple methods - 15 %	Concept of 'time work'; conservation of resources theory; labor process theory; person-organization fit; theory of planned behavior

develop a wider appreciation of the complexity of humanity (Sanusi et al., 2023).

Connecting these two themes shows that discussions about AI literacy also intersect with career exploration, although this connection may not always be explicitly articulated. For example, initiatives to cultivate AI literacy can also provide students with opportunities to gain insights into STEM-related careers (Park & Kwon, 2024; Su & Yang, 2023) and a broader range of careers that will require significant interaction with AI in the future (Hwang et al., 2022). In essence, AI literacy programs not only help develop career-relevant skills but also offer students insights into careers directly associated with AI development and use.

4.1.2. University-to-work (i.e., university students)

This career stage encompasses four distinct themes.

Using AI for career guidance and academic advising ($n = 28$). The first theme focuses on utilizing AI's predictive capabilities to offer career guidance and academic advice. This application primarily involves delivering targeted information to students regarding how specific academic programs can align with various career paths and the factors influencing positive or negative career outcomes. AI provides such insights by assessing and matching individuals' skill sets with the prerequisite skills to qualify for tertiary specializations, jobs, and careers, ultimately enhancing users' knowledge to make informed decisions about their educational and career choices (Heinisch et al., 2020). For example, AI can be used to recommend career pathways based on students' skills and occupational requirements (Mason et al., 2023; Vo et al., 2022), to help students understand the alignment of their skills with employers' demands and potential careers (José-García et al., 2022), to offer advice on suitable future job options and post-university destinations based on students' capabilities (Chen et al., 2022; Kuswanth et al., 2019), to provide insights into the skills that predict higher employability (Raman & Pramod, 2021), and to identify the academic and non-academic factors predicting career choices (Yang & Chang, 2023). These AI applications aim to provide precise career advice tailored to users' skill levels and career opportunities. This approach can be seen as a dimension of "precision education" (Yang et al., 2021), enabled by advanced educational data mining, that specifies the relationship between students' learning performance and their career decisions. A smaller sub-set of research within this theme ($n = 4$) focused on predicting students' career decisions in sectors facing recruitment challenges, such as STEM-related (Liu & Tan, 2020; Makhlouf & Mine, 2020; Yeung & Yeung, 2019) and seafaring (Fei & Lu, 2015) careers.

Beyond offering career-relevant information, other work within this theme seeks to create a platform connecting various stakeholders, including students, parents, educators, and guidance counsellors, to establish an ecosystem that facilitates interactions to support student jobseekers (El Haji & Azmani, 2020). Additionally, this theme encompasses the use of AI for academic advising, aiming to assist students in aligning their studies with their career aspirations (Obeid et al., 2020). This includes predicting academic performance to help students target their career options (Bhushan et al., 2023), offering advice to undecided students to enhance their academic success and career trajectories (Wang & Orr, 2022), and streamlining course admission processes (Usman et al., 2017; Wabwoba & Mwakondo, 2011). In contrast to the school student cohort, there is less explicit emphasis on AI supporting career exploration (for an exception, see Kurbanova & Yarovikov, 2021), possibly because it is assumed that university students have already

chosen their career paths.

Findings within this theme also highlight the broader benefits of AI, positioning it as a tool that supports human career advisors (Westman et al., 2021) and assists universities in aligning their curriculum offerings with occupational skill demands (Vo et al., 2022). Some applications of AI in career guidance also prioritize equity by targeting traditionally disadvantaged groups, such as by providing career support for female engineering students (Jang & Kyun, 2022) and addressing the accessibility gap in career counselling services in certain educational institutions (Akiba & Fraboni, 2023; Obeid et al., 2020). Overall, while questions remain about the most effective mix of AI- and human-generated career insights (Westman et al., 2021), there is evidence that students are comfortable with the use of AI for career guidance, perceiving it as supportive rather than threatening (Rodway & Schepman, 2023; Westman et al., 2021). Moreover, the use of generative AI may further enhance student engagement with AI for career guidance through more user-friendly interactions (Akiba & Fraboni, 2023).

Building students' AI literacy and views on AI's impact on future careers ($n = 11$). The second theme focuses on AI literacy, by identifying the AI skills required for specific professions and investigating students' perceptions of how AI might affect their chosen careers. This theme encompasses a range of professions, including marketing (Thontirawong & Chinchanachokchai, 2021) and water resources engineering (Habib et al., 2019), but predominantly centers around healthcare and medicine ($n = 5$). Medical students anticipate a growing application of AI in their field, leading them to believe that the development of AI literacy should occur during their studies (Al Saad et al., 2022; Teng et al., 2022). However, formal opportunities to acquire AI knowledge in their training currently appear limited (Ashrafi et al., 2014; Pucchio et al., 2022). Another study mapped the types of curriculum opportunities available to students to develop this knowledge, revealing that student-led initiatives, such as student-run clubs dedicated to innovation and technology training, often take precedence over formalized AI education (Wang et al., 2022). In contrast, certain professions provide concrete examples of relevant AI applications that can be integrated into the classroom to provide students with practical experience for their future careers (Habib et al., 2019; Thontirawong & Chinchanachokchai, 2021). Additionally, Reilly and Reeves (2023) discuss the necessity for graduates to develop transdisciplinary skills that bridge disciplinary siloes, to adapt to technology-driven demands of work.

Overall, there are diverse student perspectives regarding the impact of AI, including its generative forms, on their future careers. Some express optimism (Vicsek et al., 2024), others harbor concerns about their future employment prospects (Chan & Hu, 2023), and others hold variable views dependent on their discipline (Teng et al., 2022). Nevertheless, students consistently emphasize the importance of AI and the need for its integration into their future work (Chan & Hu, 2023). Enhanced AI literacy also has the potential to help students cultivate informed and realistic expectations regarding how technology will shape their future careers (Al Saad et al., 2022).

Examining job- or occupation-specific skills in AI-enabled workplaces ($n = 5$). The third theme examines the range of job- or occupation-specific skills that graduates require in AI-driven workplaces. This research highlights that digital skills represent one facet of a diverse skill set required. In this context, graduates will continue to apply a blend of social and technical skills, alongside core competencies in problem solving and critical thinking (Hussain et al., 2023), and the development of "new hard skills" associated with AI use (Woods et al., 2022). Within this theme, AI also serves as an analytical tool to evaluate the skills demanded by various professions (Mantai & Marrone, 2022) and to determine whether the competencies developed by higher education institutions align with industry demands (Fenton et al., 2022). Such insights should assist educational providers align their curricula with the current and future skill needs of graduates (Yamashita & Cummins, 2022).

Predicting at-risk students and student attrition using AI ($n = 5$). The final theme centers on the use of AI as a predictive tool to model the risk of student attrition. Such usage aims to pinpoint the characteristics of students who may be at risk of either dropping out of their studies or achieving sub-par academic performance (Perchinunno et al., 2021; Singh et al., 2024; Vidal et al., 2022). These predictive assessments can be tailored to cater to students from non-traditional backgrounds, including those who pursue part-time education or have caregiving responsibilities (Huo et al., 2023). Insights from predicting at-risk students and their potential attrition can empower educational institutions to implement targeted support to bolster retention and participation (Huo et al., 2023; Vidal et al., 2022). While these studies often incorporate academic performance as a key predictor of success, they also utilize personal characteristics and family circumstances (Singh et al., 2024). Some studies further identify factors contributing to successful graduation (Hannaford et al., 2021).

4.1.3. Established workers (i.e., those in employment)

This career stage comprises four overarching themes.

Understanding the career impact of AI ($n = 15$). In the first theme, scholars draw upon career theories to investigate the impact of AI on workers' career management, acquisition and refinement of career competencies, and perceptions of career paths. This literature broadly frames the use of AI as a disruptor of individuals' job roles and careers, examining both 'dark side' and 'bright side' perspectives. These studies also identify the boundary conditions, such as career resources and adaptive behaviors, that can mitigate the adverse career consequences of AI disruption. Research focusing on the dark side outcomes identifies fears of human replacement by AI, often operationalized through AI/STARA awareness measures, leading to negative outcomes for workers. For instance, Presbitero and Teng-Calleja (2022) find that call center agents who believe that AI will replace their jobs are more likely to engage in career exploration behaviors, driven by feelings of job insecurity and psychological distress. Moreover, in the service sector, employees' heightened AI/STARA awareness was directly linked to negative well-being outcomes, including job burnout, depression, and reduced career satisfaction (Brougham & Haar, 2018; Kong et al., 2021), along with weakened career competencies, albeit indirectly via reduced organizational commitment (Kong et al., 2021). Employees in repetitive and non-complex job roles are also more likely to believe that their positions will become redundant due to new technologies (Brougham & Haar, 2017).

On a more optimistic note, the fear of automation can act as a catalyst for positive career-related behaviors, including stronger intentions to invest in one's training (Innocenti & Golin, 2022), with workers generally acknowledging the importance of upskilling (Karthikeyan & Ponniah, 2019). Beyond individual actions, within the context of digitalization in the manufacturing sector, Lu et al. (2019) emphasize how organizations can nurture the career growth and employability of their employees, promoting their career sustainability, by embracing a socially responsible approach. This involves implementing strategies to enhance employee benefits, work hours, the work environment, and training. Additionally, adopting a sustainable career lens, Kong et al. (2023) demonstrate that workers who trust AI are more likely to collaborate effectively with it, contributing to their well-being, productivity, and career sustainability. The study highlights the importance of a worker's protean career orientation, characterized by agency and flexibility, as a resource that strengthens trust in AI and facilitates collaboration with the technology.

Interestingly, Zhang and Jin (2023) navigate the balance between these bright and dark side implications of AI use. They observe that hotel employees with pessimistic STAARA[2] awareness (i.e., beliefs that AI will replace human workers), are more likely to experience elevated job insecurity and job mobility compared to their counterparts with optimistic STAARA awareness (i.e., beliefs that AI will assist human workers). These negative effects are more pronounced for workers with lower-level career progression, while those with higher career progression levels are better shielded from the job insecurity associated with STAARA awareness.

In parallel, other research in this theme centers on platform workers and how the algorithmic management of gig work shapes career-related experiences. While gig workers may not perceive gig work as a long-term career (Duus et al., 2023), the concept of a 'career' is not entirely foreign to them. For instance, implementing strategies related to agency and self-management, such as self-leadership, was found to enhance the well-being of these workers and increase their intention to explore non-gig employment opportunities, which was seen as advantageous (Crayne & Brawley Newlin, 2024). However, Duggan et al. (2022) demonstrate the constraints imposed by algorithmic management on gig workers, severely limiting their capacity to develop their career competencies.

Qualitative research in this theme offers rich insights into the evolving nature of hybrid workflows introduced by AI and the broader implications for careers. In the legal profession, Armour et al. (2021) discuss how AI adoption is generating new forms of work and roles in law firms, traditionally the exclusive domain of lawyers. Multidisciplinary teams and hybrid roles, including legal engineers, legal analytics specialists, and e-Disclosure specialists, are gaining prominence. The accompanying changes in tasks, professional jurisdictions, occupational identities, and organizational structures are also re-shaping career pathways. For example, junior lawyers are facing shifts in career opportunities through fewer partner-track positions, while the demand for career paths for staff without legal backgrounds is increasing. In contrast, within the audio mastering profession, Birtchnell (2018) provides a deep exploration of how these creative professionals integrate AI technology into their workflows, leveraging its advantages while expanding their unique human capabilities and re-shaping their ideas for career development. Similarly, in the visual arts context, Johnston and Lane (2019) show how artists navigate technological disruptions by developing their complex cognitive and creative skills within evolving creative mediums and environments to foster successful artistic careers. Overall, this theme demonstrates the varied impact of AI on workers' careers through both the challenges and opportunities it presents, while highlighting the importance of adaptive behaviors and innovative career strategies.

AI predictions of workers' performance and career trajectories ($n = 12$). The second theme discusses the use of AI as a tool to assess workers' performance, often to predict their future career trajectories based on quantifiable performance indicators. This includes the prediction of academic researchers' performance (Zhou et al., 2022), impact (Ayaz et al., 2018; Zuo & Zhao, 2021), and influence (Kumar et al., 2023), via the analysis of metrics such as citation scores (Zafar et al., 2023), impact factors, and the nature of collaborative research networks (Costa, 2014; Zuo & Zhao, 2021). Such predictions can facilitate recruitment and promotion of staff (Zhou et al., 2022), help researchers in determining their research direction (Zafar et al., 2023), support their career advancement (Zuo & Zhao, 2021), more fairly allocate grant money (Ebadi & Schiffauerova, 2016), and evaluate the impact of development opportunities (Ke et al., 2022). Beyond academia, these machine learning approaches are also harnessed to predict the performance of lawyers (Almeida Lopes et al., 2023; Lopes et al., 2018) and identify the factors leading to high performance for sales representatives (Glackin & Adivar, 2023), providing a foundation for tailoring career plans and enhancing performance.

AI-enhanced career guidance ($n = 8$). The third theme centers on using AI for predicting career paths and providing guidance for learning and skill acquisition, particularly in the context of career guidance. This theme demonstrates that, compared to traditional approaches, machine learning can enhance the accuracy of career predictions based on individuals' vocational interests (Song et al., 2024). Given the important role of career choices in influencing downstream outcomes such as job satisfaction and turnover (Song et al., 2024), improving the precision of these predictions can offer many benefits for workers and organizations. Offering an equity dimension to such predictive capabilities, Hickey et al. (2022) leverage machine learning to identify the career-related decisions and behaviors of women who have attained leadership positions in the 'architecture, engineering, and construction' industry. Their identification of the importance of pursuing advanced degrees, expanding professional networks, and changing employers offers insights for early- and mid-career women regarding the steps that have led to others' career progression.

Other studies in this theme focus on the development and use of AI-powered recommender systems to assist individuals navigate the vast array of potential job opportunities (Parida et al., 2022) and learning pathways (Ahmad et al., 2023; Kokkodis & Ipeirotis, 2021). These systems filter and target job and learning options based on individual competencies and desired career paths (Ahmad et al., 2023; Parida et al., 2022). They also account for contextual factors such as company recruitment needs (Parida et al., 2022) and current and future market trends in skill demands (Kokkodis & Ipeirotis, 2021). Additionally, chatbot interfaces are proposed to

[2] In this study, the operationalization of STAARA includes 'automation' technology (STA**A**RA: smart technology, artificial intelligence, **automation**, robotics, and algorithms).

facilitate digital counselling in the process of course selection for skill development (Ahmad et al., 2023). Some research within this theme also adopts an organizational perspective, focusing on the development of models that optimize staff training in alignment with job requirements and career paths (Karamalak et al., 2014).

However, it is important to note that studies within this theme emphasize the critical aspects of AI model selection and specification for career guidance purposes, which can sometimes be overlooked. For instance, when considering recommender systems for skill development opportunities, Kokkodis and Ipeirotis (2021) discuss the limitations of traditional or mainstream recommender systems, which recommend products and services based on past behavioral patterns. Such systems may hinder the development of new skills, reinforce past sub-optimal decisions made by others (which inform the recommendations), and fail to leverage individuals' diverse learning abilities and likelihood of skill acquisition. Additionally, in terms of predicting career choices, Song et al. (2024) highlight that machine learning models may exhibit reduced accuracy in certain situations, particularly when predicting career choices in less popular job categories. Consequently, AI may not always be the optimal choice for all career-related predictive tasks. Relatedly, questions of model transparency are explored by Gedrimiene et al. (2023), revealing that trust in AI-generated career advice varies among users. Accuracy is shown to be a strong predictor of intention to follow such career recommendations, with age playing a role as older users place greater value on transparency and exhibit a more critical stance toward the recommendations.

This theme illuminates how AI can be used in forecasting career paths and guiding learning and skill acquisition for career development. While AI offers advantages toward meeting these ends, careful consideration of model selection, transparency, and context-specific suitability is paramount for its effective use in career guidance and prediction.

Skill needs due to AI use ($n = 4$). The final theme examines the evolving skill needs resulting from the increasing organizational use of AI. This research highlights AI's dual role as both a powerful analytical tool and a disruptor of work-related skills. With access to extensive occupational datasets, most of these studies harness AI, including machine learning and natural language processing, as analytical tools to identify shifts in skill requirements driven by the technology's impact on various professions.

These analyses shed light on a range of aspects, including the skill sets demanded of early career researchers and the prerequisites for entry-level research positions (Maer-Matei et al., 2019). They also reveal emerging skill patterns among platform workers, spanning specialized/general and creative/technical roles (Ivanovic et al., 2023). The impact of technology-driven skill changes also necessitates a commitment to lifelong learning, to mitigate unemployment in occupations characterized by 'harder' (viz. technical) skills and to foster upward career trajectories in 'softer' (viz. socioemotional) skill-based occupations (Schultheiss & Backes-Gellner, 2023). However, there are concerning findings in certain occupational contexts. For example, Pshenichnykh and Novi (2023) highlight a potential gap in the preparedness of hospitality and tourism workers for the digital competencies required in the future. Survey results indicate that these workers self-report their lowest skill levels being those associated with working with AI. Further, they believe that these skills will not be demanded in the future. This perception contrasts with an analysis of job requirements in the sector, which already demonstrates a demand for such digital capabilities. Such work underscores the importance of alignment between workers' perceptions of skill demands and the actual requirements of their industries, to support their careers in AI-enabled workplaces.

4.1.4. Cross-cutting themes: evaluating the discussion of AI across career stages

So far, our presentation of findings has helped pinpoint how AI influences key milestones *within* career stages. In the following section, we synthesize the similarities and differences observed in the themes *across* career stages.

AI used for career guidance and prediction emerges as a recurrent theme across all career stages. This shows how AI's predictive capabilities are harnessed in similar ways to predict educational and career trajectories, serving as a tool to alleviate stress and uncertainty associated with significant life decisions. By providing evidence-based guidance, AI can empower individuals to make informed choices. However, the discussions also touch upon issues concerning the predictive outcomes of machine learning approaches, such as concerns regarding lack of explainability, transparency, and trustworthiness. While AI's role in *predicting* career decisions is explored in depth, its potential to assist individuals in *exploring* a wide range of career paths, including through access to virtual mentors (Nye et al., 2021), receives less emphasis across all career stages. However, such guidance would seem crucial, especially during the formative school and university stages, as well as in the established worker stage when individuals contemplate career transitions or face unemployment.

The *examination of AI literacy* and the cultivation of AI-specific knowledge are positioned as important components of a required future skill set. This emerges as a consistent theme in the school and university career stages. However, a key distinction lies in the effectiveness of skill development efforts and outcomes in these stages. The primary focus for school students is on building their general knowledge of AI/machine learning, encompassing basic technical knowledge and an awareness of the technology's implications, including its impact on jobs. While the pedagogical approaches to achieve this vary (Su & Yang, 2023), a consensus is forming regarding the types of AI-relevant knowledge that should be developed during this career stage. For university students, the focus is more on students' perceptions of how AI would impact their careers and their views on how AI-relevant knowledge is integrated into their curricula. A key finding is that while students view AI-relevant knowledge as important, universities—both from students' perceptions and objective evaluations of their curricula—are not currently providing this occupation-specific AI knowledge effectively. In contrast, AI literacy was not a clear theme at the established worker stage. This perhaps suggests an underlying assumption that such skill development is unnecessary at this stage, either because it is assumed to have been acquired in an earlier career stage, or because it is overlooked despite its importance. It may also be the case that emerging research on workers' AI literacy is not yet being integrated within the careers field (e.g., Cetindamar et al., 2024). Although the exploration of future skill requirements briefly addresses this issue, there remains a lack of comprehensive and definitive insights into AI-related skills for established workers.

Notwithstanding the above, discussions concerning job- or occupation-specific skills resulting from AI use emerged as a recurring theme across both university student and established worker stages. The university student cohort often connected these insights about

skills with higher education institutions, viewing them as the primary entities responsible for adapting their curricula. In contrast, the established worker cohort generally did not attribute these skill-based insights to any specific entity for action, be it employers, government bodies, or industry associations. As highlighted by Ahmad et al. (2023), the decentralization of skill development beyond secondary and higher education implies that the onus falls upon workers to comprehend these skill requirements and take measures to develop them.

The *use of AI to identify vulnerable students or workers*, to target support interventions, was unevenly discussed. Despite needing to support school students' academic performance and to assist workers facing technological disruptions, it is largely research in the university career stage that examines the use of AI to identify tertiary students at risk of study non-completion. Exceptions exist (see Alghamdi & Rahman, 2023; Schultheiss & Backes-Gellner, 2023), but typically, school students facing similar performance risks and established workers potentially navigating significant changes in task and skill requirements have not received commensurate attention in this regard. Instead, in the established worker stage, AI/STARA awareness often serves as self-reported proxies for workers' concerns about AI replacing them, without necessarily evidence of actual risks of replacement. This observation also underscores the dearth of research on AI literacy in the established worker career stage. Some authors (e.g., Brougham & Haar, 2017; Pshenichnykh & Novi, 2023) suggest that this lack of attention to AI literacy could result in workers possessing limited or inaccurate perceptions of how AI will genuinely impact their jobs.

Overall, research conducted in the established worker career stage reveals the already *varied and conflicting effects of AI on careers*. Our review illustrates that while the integration of AI into workplaces can potentially bolster workers' sustainable careers (e.g., Kong et al., 2023), it may also have adverse effects on their productivity and well-being (e.g., Brougham & Haar, 2018). Even within individual-level studies, there is the juxtaposition of employers' demands for new forms of digital skills and workers' beliefs that AI-relevant skills may not be crucial for their future work (e.g., Pshenichnykh & Novi, 2023). These outcomes shed light on the inherent challenges associated with building workers' AI literacy and cultivating occupation-specific knowledge of AI. Understanding the effects of this technology and workers' responses to it remains an ongoing process and, as our review shows, the exact nature of these effects will vary across occupations.

5. Barriers and enablers of utilizing AI within and across career stages

Researchers' discussions of the barriers and enablers of AI use often revolve around the functionality of AI models for career purposes. As these barriers and enablers emerged across the career stages, we consolidate their discussion here.

A frequently cited enabler of AI use in the career domain is the accessibility of vast datasets necessary for their operation. This accessibility has been driven, for instance, by the increasing adoption of learning analytics and data collection practices by universities (Yang et al., 2021). Effective integration of datasets is also facilitated through intra-organizational collaboration between relevant parties, such as data analytics staff and career/academic advisors (Wang & Orr, 2022). Research also indicates that key end users, including university students, generally hold positive attitudes toward the use of AI for career-related advice, which can facilitate its adoption (Rodway & Schepman, 2023; Westman et al., 2021).

On the contrary, barriers to AI adoption for career purposes are predominantly associated with the quality of data used, the nature of the models, and their resulting outputs. While these issues are well-known challenges in the broader application of AI, they become particularly critical in the sensitive context of career guidance, where the use of AI is potentially shaping a crucial aspect of people's lives. A prominent barrier is the variability in data quality, which affects the accuracy of the models and their resulting outputs (Assayed et al., 2023; Hannaford et al., 2021; Hernandez & Atienza, 2021; Jang & Kyun, 2022; José-García et al., 2022; Mason et al., 2023; Obeid et al., 2020). The accuracy concern is highlighted by Song et al. (2024), who demonstrate that while machine learning models can enhance predictive accuracy in some scenarios, they may perform sub-optimally in others, such as predicting career choices for occupations with relatively few workers. Furthermore, there is a risk of biased outcomes when using data with specific indicators, such as geographic location (Carson et al., 1999), potentially reinforcing existing societal inequalities by offering more opportunities to advantaged groups and fewer to disadvantaged ones. Despite the volume of data available, AI models may still fail to capture every indicator relevant to predicting complex phenomena like career choices, including students' (often changing) values and preferences (Carson et al., 1999). Cumulatively, these issues can negatively impact the predictive accuracy of AI outputs (Alghamdi & Rahman, 2023; Usman et al., 2017), potentially leading to erroneous conclusions and advice for end users.

The opaque, 'black box' nature of neural network models renders their computations unknowable and so unexplainable to users (Wabwoba & Mwakondo, 2011; Yang et al., 2021; Yang & Chang, 2023). This, combined with accuracy issues, can affect users' trust in these systems, with the extent of this trust varying depending on user characteristics such as age (Gedrimiene et al., 2023). Concerns related to users' data privacy and ensuring data security are also discussed (Jang & Kyun, 2022; Su & Yang, 2023; Westman et al., 2021). These issues merit further investigation given that educational data mining, which can underpin the use of AI for career-related functions, often draws data from users' interactions with online learning platforms. This raises questions about whether users are genuinely aware of, or meaningfully consent to, their data being utilized or repurposed for such predictive outcomes.

Beyond data and modeling concerns, the need to recruit and train educators and career counsellors to develop and deliver AI literacy content or incorporate AI outputs into their guidance services are also acknowledged as significant yet unresolved issues (Habib et al., 2019; Hussain et al., 2023; Ng et al., 2023; Pucchio et al., 2022; Woods et al., 2022). Finally, research in the established worker career stage shows that organizational AI use can lead to adverse outcomes for employees, acting as a potential barrier to sustainable careers.

6. Impact of AI use on individuals' career competencies within and across career stages

We interrogate our findings on career competencies by adopting Akkermans et al.'s (2013) three-pronged model of career competencies. *Reflective competencies* (knowing-why) involve adopting a long-term perspective to one's career and reflecting on what motivates oneself to engage in a particular career and identifying the skills and qualities needed in that career. *Communicative competencies* (knowing-whom) involve engagement with others, such as through networking, and the ability to promote one's skills to relevant labor markets. Finally, *behavioral competencies* (knowing-how) describe the intentional effort to influence one's career by searching for appropriate career opportunities and exerting control over one's career by setting goals and taking necessary steps to achieve them.

6.1. School (i.e., primary and high school students)

In the school student career stage, our findings show that the use of AI for career guidance can support the development of *reflective competencies*. AI offers insights into the nature of various careers and the educational pathways required to pursue them (e.g., Assayed et al., 2023). Some studies also demonstrate how AI supports self-assessment, providing valuable information for career planning (Singla et al., 2022). For *communicative competencies*, AI could assist career counsellors in improving their feedback and in enabling on-demand self-service access to career advice for end users, through platforms like chatbots. Discussions surrounding AI literacy contribute to the development of *behavioral competencies* by developing AI-specific and related skills that enhance career adaptability.

6.2. University-to-work (i.e., university students)

In the university student career stage, the use of AI to predict career choices and trajectories helps develop their *reflective competencies*, by identifying the determining factors behind various career paths. However, there was less emphasis on AI used for exploratory purposes, compared to the school student stage. This difference might be attributed to an implicit assumption that university students have already undertaken sufficient career exploration and self-reflection prior to this stage. Nevertheless, opportunities for reflection remain crucial for university students, as a lack of such opportunities could increase the risk of students dropping out when they realize that their chosen course of study does not align with their expectations. There is potential to expand the use of AI to support these students' career reflections and assist those who become uncertain about their career trajectory.

Like the school student cohort, the use of AI for on-demand career guidance can help university students engage with a trusted, albeit non-human, source for career advice, supporting *communicative competencies*. This extends to initiatives that harness AI to create a digital ecosystem for career guidance (El Haji & Azmani, 2020). However, as highlighted by Kuijpers et al. (2011), career guidance is fundamentally a dialogue, without which the development of relevant career competencies is constrained. This raises questions about the extent to which AI-generated career guidance should be autonomously conveyed to students or instead be used as a tool alongside human career advisors, who remain the primary source for interpreting these insights. It is worth noting that only one paper in this career stage utilized AI to support human mentoring opportunities for students (Jang & Kyun, 2022), a pathway that would seem fruitful for leveraging a human- *and* technology-centered solution.

Additionally, the use of AI for career guidance and prediction can assist students in planning the specific steps needed to achieve their career goals, developing their *behavioral competencies*. For instance, Fulk et al. (2022) demonstrate how AI could aid students in interview preparation, promoting the development of self-profiling skills within this competency. However, further work is needed to integrate AI literacy into relevant curricula to advance behavioral competency development, especially adaptability, at this career stage.

6.3. Established workers (i.e., those in employment)

Our findings in the established worker stage reveal a spectrum of implications for the ongoing development of workers' career competencies in the context of AI use. Once again, AI's ability to provide insights into the factors influencing career choices is valuable for helping established workers reflect on their skills and compare them to the requirements of their desired careers, thereby supporting *reflective competencies*. While engaging with AI for this purpose could be seen as an emerging form of human-AI advising, which could also contribute to *communicative competencies*, striking the right balance between AI and human-based advising remains a challenge that requires further exploration.

Moreover, the lack of a clear emphasis on AI literacy and the development of occupation-specific, AI-relevant skills hinders the enhancement of *behavioral competencies* among established workers. Without a well-defined understanding of these skill requirements, workers may find it challenging to plan and execute necessary actions to remain competitive in the labor market. It is also arguable that such re- and upskilling should not fall solely to workers themselves, and requires interventions from relevant educational bodies or organizations themselves to support such endeavors. The narrow focus on service-based occupations further limits our knowledge of how a diverse range of workers' career opportunities will be influenced by AI.

7. Future research directions

Our review provides valuable insights into the current applications of AI within career and vocational domains. Like the impact of various technologies throughout history, the adoption of AI in these domains entails both benefits and costs. However, we assert that

there is currently insufficient critical attention directed to understanding the limitations and risks associated with AI's use in these domains. Challenges associated with AI have been discussed in the broader HRM field, prompting governments to take steps to influence and regulate its use in high-risk contexts such as recruitment (e.g., in Australia; Department of Industry, Science and Resources, 2024; and in the European Union; European Commission, 2024). However, there is a noticeable absence of a wider conversation about how these challenges may impact people's *careers* more broadly.

To foster a balanced discourse, our research agenda outlines ways to explore how AI can be harnessed to support careers across the lifespan, while also presenting avenues for critique to identify and mitigate potential harms from this use. This is crucial because career choices not only influence individuals' livelihoods at a specific point in time but can have repercussions across their entire lifetime. These choices can ultimately shape or diminish opportunities for meaningful and dignified work, adequate income, the development of one's capabilities for a good life, and even break cycles of intergenerational disadvantage.

To develop and organize our future research suggestions, we adopt a sustainable career lens (De Vos et al., 2020). This perspective recognizes careers as dynamic, unfolding journeys over time, which aligns with our stage-based approach to reviewing the literature. By embracing this theoretical lens, we formulate research directions that encompass issues across the lifespan of individuals' careers. Sustainable careers refer to "sequences of career experiences reflected through a variety of patterns of continuity over time, thereby crossing several social spaces, characterized by individual agency, herewith providing meaning to the individual" (Van der Heijden & De Vos, 2015, p. 7). This concept comprises three dimensions: a) the person dimension, reflecting the idea that individuals are central agents actively taking career-related decisions and actions; b) the temporal dimension, conceptualizing careers as evolving, rather than static; and c) the context dimension, which recognizes that career-related actions and choices are influenced by a range of factors in personal, organizational, and societal spheres. Indicators of a sustainable career include feelings of career-related *happiness* derived from experiencing success, meaning, or joy in one's work, experiences of physical and mental *health*, and *productivity* resulting from performing well and developing employability (De Vos et al., 2020).

By applying the notion of sustainability to our future research agenda, we generate recommendations that prioritize social justice, a fundamental principle within the career development field (Lent & Brown, 2013). This approach encompasses ethical considerations in the deployment of AI across its various applications, ensuring a balanced and comprehensive perspective on the impact of AI on individuals' careers.

7.1. Happiness in careers

7.1.1. Person dimension: ensuring equity in AI-enabled career prediction

AI models operate by generating predictions of future outcomes based on past data. It is well-established that such processes can result in biased outputs, for example, by replicating systemic societal injustices against minority groups. This issue is exemplified in the use of AI for predictive policing, where it can result in the over-policing of certain racial groups, and in job recruitment, where it can perpetuate occupational gender imbalances. These concerns are highly relevant in the context of AI use in the career domain and need much greater research attention. Indeed, there have been concerns raised about the use of career predictors that might inadvertently reinforce discrimination (Carson, 1999). Such predictors could potentially steer individuals away from certain careers that have historically lacked diverse representation, thereby perpetuating past career barriers. However, it is worth noting that our literature sample did not extensively delve into these issues, with some exceptions (see Gedrimiene et al., 2023; Song et al., 2024). To address these challenges, the future development of AI models for career prediction and guidance should adopt a proactive approach. Rather than relying on readily available datasets, there should be a concerted effort to enhance the ethicality of training datasets, as demonstrated in the healthcare context (Gaonkar et al., 2023). Moreover, there is a need to implement strategies that yield better outcomes for under-represented groups, with the goal of rebalancing and diversifying occupational demographic profiles (Song et al., 2024).

7.1.2. Person dimension: accounting for, and expanding, hopes and aspirations

An open question is how well AI models consider factors that may be challenging to quantify or capture, but that are crucial for determining subjective career happiness. While our review shows that predictive datasets can encompass a variety of academic and non-academic variables, including indicators of career interests (see Song et al., 2024), it is essential to recognize that a multitude of factors, subject to change over a person's life, play a role in shaping career choices. For instance, should or could AI models consider factors that are generally considered beyond the scope of vocational 'interests', such as a person's hopes, aspirations, and values, in modeling individual career choices? Aspirations, for example, are complex, forward-looking, driven by both conscious and unconscious motivations, and commit people to certain outcomes (Hart, 2016). Such factors could be significant, especially when modeling career choices for vulnerable groups, university students at risk of attrition, and established workers in precarious employment. For example, du Plessis and Ahmed (2020) discuss how Black youth in South Africa continue to face unequal educational opportunities, making their own and their parents' aspirations critical influences on future careers. Their qualitative research reveals how cultural values, such as leveraging their careers to support their families and communities, are significant drivers of students' career aspirations. It is unclear how current approaches to AI modeling of career choices would account for these predictors or assist marginalized groups to make career decisions aligned with their aspirations and cultural values.

However, individuals from such backgrounds are less likely to have substantial access to human career counselling services. Consequently, AI-supported career services could offer valuable guidance where none currently exists. This underscores the importance of future research into how and whether AI, along with its associated datasets and predictors, can provide meaningful career guidance that genuinely empowers individuals from disadvantaged backgrounds through their career choices (Restubog et al., 2021).

Connected to the context dimension of sustainable careers, individuals' interests and their ability to form aspirations can be constrained by their socio-economic circumstances (Hart, 2016). Therefore, throughout their lives, individuals may require support to break free from shackled aspirations that limit their potential career options and the actions associated with those choices. In this context, Nye et al. (2021) demonstrate how AI can grant students access to virtual mentors through natural language processing interactions that can enhance the availability of career mentors and services. Future efforts might explore ways to expand such initiatives to improve the accessibility of human-centered, but AI-enabled, career guidance for a broader range of individuals across different stages of their careers. This could encompass the use of immersive technologies like virtual reality, allowing individuals to 'see and experience' the realities of specific career paths.

7.1.3. Person dimension: facilitating exposure to emerging and niche occupations

The landscape of work is undergoing significant transformations, marked by the emergence of digitally-enabled career paths such as digital nomads and influencers (Aroles et al., 2023). Alongside these changes, there exist specialized and niche occupations that may align well with an individual's interests and aspirations. We note, however, that individuals may not typically encounter these career options early in their decision-making processes. These evolving forms of work are directly relevant to the utilization of AI in career prediction and guidance, as machine learning systems can exhibit reduced predictive accuracy for job categories with a limited number of individuals (Song et al., 2024). This scenario raises questions about the efficacy of AI systems in directing individuals toward unique, less traditional career paths. In cases where AI systems fall short in this regard, it prompts a consideration of potential technical solutions to overcome these limitations or whether users should be informed about these constraints through explanatory mechanisms.

7.1.4. Person dimension: supporting later-in-life career decisions

Our review highlights a notable gap in the utilization of AI across career stages, with limited representation beyond the relatively young, established worker stage. This omission overlooks a significant and expanding demographic consisting of later-career and retired workers. The latter group may still engage in various forms of paid or unpaid work to maintain a sense of purpose and social connections. It is essential to recognize that the attitudes and requirements of individuals in this career stage differ significantly from those at earlier stages. For instance, drawing upon the theory of work adjustment which emphasizes the establishment of a complementary relationship between an individual and their work environment, Harper and Shoffner (2004) highlight the importance of assisting retirees in identifying activities that can compensate for the losses experienced when transitioning away from their primary career. This process often involves adapting existing skills rather than necessarily acquiring new ones. Exploring how AI systems can effectively address the unique career needs of individuals in these later-career stages would offer insights to broaden the scope of AI's benefits across the full spectrum of the career lifespan.

7.1.5. Context and temporal dimensions: incorporating the role of serendipity in careers across the lifespan

A sustainable career lens positions unplanned events as serendipitous experiences that can be leveraged for new career discoveries (De Vos et al., 2020). Other careers theories, such as happenstance learning theory (Krumboltz et al., 2013) and the planned happenstance perspective (Urbanaviciute et al., 2019), also emphasize the significance of serendipity in shaping career development. However, AI-based recommender systems are known to grapple with a serendipity problem (De Gemmis et al., 2015). These systems often provide recommendations based on algorithms built on past actions and experiences or offer insights that individuals could have developed independently (Kotkov et al., 2023). This situation creates a potential conflict between a theoretically important driver of career outcomes and the capabilities of AI recommender systems, prompting the question of whether the extensive integration of AI in the careers domain might lead to an overreliance on prediction while diminishing the roles of chance, serendipity, and happenstance in guiding individuals' careers. This is particularly relevant as organizational turbulence increasingly drives career shocks that can result in significant career changes.

Future research could concentrate on designing AI systems that explicitly incorporate serendipitous events (as proposed by De Gemmis et al., 2015), while also shedding light on how individuals adapt to unforeseen career events and the resulting outcomes. From the perspective of individuals experiencing career shifts and shocks, further investigation could explore the optimal configurations of human (e.g., counsellors, mentors, trusted advisors) and AI-generated career advice to navigate such events effectively.

7.2. Healthiness in careers

7.2.1. Person dimension: looking beyond knowledge workers

Our review also highlights an emphasis on the use of AI to guide the careers of predominantly knowledge-based workers, a limitation acknowledged in the broader field of careers scholarship (Koekemoer et al., 2019). It is worth noting that not everyone's interests and aptitudes align with knowledge-based work, and many individuals are attracted to vocational and blue-collar careers. The nature of our literature sample, which primarily covers the transition from university to the workforce, tends to marginalize career paths that do not require higher education. Additionally, our sample of established worker literature was primarily dominated by service- and knowledge-based occupations. Future research could place a concerted effort into curating datasets and designing AI systems that provide individuals with information and guidance related to vocational careers, ensuring that AI-driven career guidance is inclusive of a wider range of career pathways.

7.2.2. Context and temporal dimensions: predicting stressors

The predictive capabilities of AI could offer insights into the nature and intensity of stressors associated with different career

pathways. Throughout their lives, individuals are likely to have varying thresholds for the types of work demands they are willing and able to manage, particularly as their family and life commitments evolve (De Vos et al., 2020). For example, the job demands a recent graduate is willing to accommodate may substantially differ from those of a mid-career professional with caregiving responsibilities. These personal thresholds can serve as predictors of an individual's alignment with specific career paths over time, based on the categorization of jobs according to their physical and mental demands. Furthermore, if AI systems can be harnessed to predict which career pathways are more or less likely to result in burnout, this would provide valuable insights for workers, employers, and policymakers. Such predictions can help identify areas where interventions are needed to reduce burnout rates and enhance the health of individuals in careers with high stressors.

7.3. Productivity in careers

7.3.1. Person dimension: evaluating portability of performance data

AI is already being deployed in organizations for the purpose of monitoring and evaluating employee performance, a trend that is evident in the research reviewed in the established worker career stage, with a particular focus on academic researchers. While electronic performance monitoring can be perceived negatively and seen as a form of surveillance by workers (Ravid et al., 2023), there is room for future research to explore whether and how access to such performance metrics could support the career development of employees. For instance, if these metrics can be easily transferred and applied across different employers, they could then serve as a type of performance portfolio. Such portfolios would enable workers to showcase their skills and achievements throughout their careers in a tangible and quantifiable manner. As individuals accumulate skills over time, the precise nature and extent of these skills may not always be readily apparent. However, AI systems have the potential to offer a means of capturing and presenting these skills comprehensively. This approach should also emphasize the importance of co-designing AI-enabled performance monitoring systems to benefit both employees and employers, fostering a collaborative effort to enhance career development and work performance.

7.3.2. Context dimension: considering AI literacy within and across career stages

A prominent theme that emerges from our review is the lack of consensus regarding when and how best to develop AI literacy skills. The sustainable career lens highlights that individuals are not the sole actors in their career journeys; they often need support from various stakeholders, including their employers, government bodies, and personal networks. The challenge of fostering AI literacy throughout one's career reflects this multi-stakeholder requirement, as there are emerging initiatives at school and university levels to nurture both general and occupation-specific AI literacy. However, the picture is far less clear regarding how established workers can continue to cultivate these skills once they have moved beyond formal educational settings. There is a pressing need for future research to develop strategies to foster AI literacy across different career stages and career paths. This research should engage multiple stakeholders, including workers, employers, higher education institutions, relevant professional associations, and government bodies, to create comprehensive and effective approaches to building AI literacy across the career spectrum.

8. Conclusion

AI systems are redefining the landscape of career choices, development, and vocational experiences. Our interdisciplinary review reveals the growing impact of AI across various career stages, highlighting its role in guiding career decisions through predictive capabilities, altering the skill sets demanded of workers, and giving rise to workers' apprehensions regarding the future of careers in an AI-driven world. Guided by a focus on sustainability and equity, our future research agenda offers recommendations aimed at directing AI development toward cultivating sustainable careers across the lifespan. Alongside empowering practitioners to harness the potential of AI to support people's careers, these insights will help individuals make informed career decisions in the age of AI, guide AI developers in the responsible design of career-oriented AI systems, and provide educators and policymakers with the tools to actively steer AI adoption toward positive and equitable outcomes. Our aim is to support AI development and deployment that fosters career choices and experiences that are not only productive but also imbued with fulfilment and healthiness, leading to a future where AI serves as an ally in building sustainable careers.

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CRediT authorship contribution statement

Sarah Bankins: Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Stefan Jooss:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Simon Lloyd D. Restubog:** Writing – review & editing, Writing – original draft, Supervision, Project administration. **Mauricio Marrone:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Anna Carmella Ocampo:** Writing – review & editing, Conceptualization. **Mindy Shoss:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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